

1. Consider a paging system in which M1 has a capacity of 5 pages. The execution of a program requires 6 distinct pages 1,2,3,4,5,6 and 7. The page address stream formed is 6 0-1-3-0-3-0-0-2-3-0-3-2-7-2-0-1-7-5-1. Visualize the action of FIFO and LRU

- **Page Frame Capacity (M1):** 5 pages
- **Page Address Stream:** 6, 0, 1, 3, 0, 3, 0, 0, 2, 3, 0, 3, 2, 7, 2, 0, 1, 7, 5, 1

1. FIFO (First-In-First-Out):

- FIFO replaces the oldest page in the memory whenever a new page needs to be loaded.

Address	Frame (FIFO)	Page Fault
6	6	Fault
0	6, 0	Fault
1	6, 0, 1	Fault
3	6, 0, 1, 3	Fault
0	6, 0, 1, 3	No Fault
3	6, 0, 1, 3	No Fault
0	6, 0, 1, 3	No Fault
0	6, 0, 1, 3	No Fault
2	6, 0, 1, 3, 2	Fault
3	6, 0, 1, 3, 2	No Fault
0	6, 0, 1, 3, 2	No Fault
3	6, 0, 1, 3, 2	No Fault
2	6, 0, 1, 3, 2	No Fault

Address	Frame (FIFO)	Page Fault
7	0, 1, 3, 2, 7	Fault
2	0, 1, 3, 2, 7	No Fault
0	0, 1, 3, 2, 7	No Fault
1	0, 1, 3, 2, 7	No Fault
7	0, 1, 3, 2, 7	No Fault
5	1, 3, 2, 7, 5	Fault
1	1, 3, 2, 7, 5	No Fault

FIFO Total Page Faults: 10

2. LRU (Least Recently Used):

- LRU replaces the least recently used page in the memory whenever a new page needs to be loaded.

Address	Frame (LRU)	Page Fault
6	6	Fault
0	6, 0	Fault
1	6, 0, 1	Fault
3	6, 0, 1, 3	Fault
0	6, 0, 1, 3	No Fault
3	6, 0, 1, 3	No Fault
0	6, 0, 1, 3	No Fault
0	6, 0, 1, 3	No Fault

Address	Frame (LRU)	Page Fault
2	6, 0, 1, 3, 2	Fault
3	6, 0, 1, 3, 2	No Fault
0	6, 0, 1, 3, 2	No Fault
3	6, 0, 1, 3, 2	No Fault
2	6, 0, 1, 3, 2	No Fault
7	0, 1, 3, 2, 7	Fault
2	0, 1, 3, 2, 7	No Fault
0	0, 1, 3, 2, 7	No Fault
1	0, 1, 3, 2, 7	No Fault
7	0, 1, 3, 2, 7	No Fault
5	0, 1, 2, 7, 5	Fault
1	0, 1, 2, 7, 5	No Fault

LRU Total Page Faults: 9

Comparison of FIFO vs. LRU:

- **FIFO Total Page Faults: 10**
- **LRU Total Page Faults: 9**